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An integrated Lexicographic goal programming and dynamic satisfaction function model for effective nurse scheduling

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ABSTRACT

The nurse-scheduling is a common problem in hospitals. The problem requires a timetable considering hospital regulations and nurses' demands and preferences. This study proposes an integrated Lexicographic goal programming and dynamic satisfaction function model for solving nurse-scheduling problems. The proposed model provides the head nurses with a balanced and fair automated schedule. The model considers two aspects related to hospital regulation constraints. First, the dynamic aspect allows the decision-makers to integrate their judgment to the changing circumstances caused by nurse absenteeism. Second, the satisfaction aspect considers the decision maker's preferences in assigning nurses' surplus or shortages to the shifts. The model's primary goal is to give nurses a fair and balanced schedule with an evenly distributed workload. A fair and balanced schedule helps maintain nurses' welfare and provide quality patient service. We present a case study in an intensive care unit to demonstrate the effectiveness of our approach compared to the conventional goal programming method.

1. Introduction

The management of healthcare systems is complex [1,2]. It requires the management of rare resources to meet the growing needs of patients [3]. Among these resources, there is the nursing staff which has been the center of interest of several researchers. The nurse scheduling problem is an NP-hard problem [4], it improves the operational efficiency of hospital services through the optimal use of limited resources, which will have a positive impact on the well-being of nurses and patients [5]. The nurse scheduling problem is an important problem and difficult to solve due to different constraints (labor legislation, contractual agreements, nurses' preferences). These constraints are specific to each hospital. The increased number of constraints and nurses makes the planning process more difficult [4]. The nurse scheduling problem involves the assignment of nurses to work schedules while taking into consideration hard and soft constraints. The hard constraints are related to hospital regulations which must be satisfied to obtain a realizable schedule [6]. The soft constraints are related to non-obligatory and preferable requests. These constraints may not be satisfied such as nurses' preferences. The objective of their insertion is to improve the quality of the feasible schedule [6]. Many head nurses still create schedules manually. Manual resolution of a real case requires much time and generates an increased administrative workload [7]. Manual preparation followed by optimal adjustment of the schedule taking into account staff requests and hospital policies requires at least one day of work on the part of the head nurse [8]. Automating schedules through nurse scheduling problem improves this process, by assigning nurses to shifts and days based on their preferences and the needs of the hospital [9]. Generating an optimal schedule for nurses improves hospital efficiency and avoids problems of understaffing or overstaffing [5]. It is very important to manage the disruption caused by absenteeism but when the timetable is prepared manually it becomes too difficult to resolve [10]. Generally, the rate of unscheduled absences is between 5 and 10%. And compared to other sectors, the highest absenteeism rates (10.7%) are in emergency services and hospital healthcare [10]. In our case, we offer an automated method to manage unplanned absences and help the decision-maker quickly resolve disruptions. Nurse schedules are usually prepared monthly or weekly. Their preparation is difficult and complex because it is based on multiple conflicting goals. To achieve the satisfaction of the hospital's demand during the planning horizon, we need to maximize the nurses' satisfaction and distribute the workload equitably. The Nurse Scheduling Problem is particularly challenging because Healthcare institutions must consider skills requirements, work regulation compliance, nurse shortage/absence, and nurses preferences. Moreover, it is essential to evenly balance the workload among the nurses and satisfy their preferences. A schedule that satisfies the nurses has a positive impact on the working environment. A quality schedule can satisfy nurses and generate an effective workforce [11]. A critical measure for the quality of a nurse's schedule is the perceived fairness or balance of nurses' workload [11]. The number of nurses scheduled per shift varies according to the hospital and depends on several

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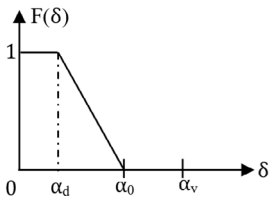
factors such as annual budgets and estimated patient volume and/or acuity [12]. The reasons that make nurse scheduling multi-objective, dynamic, and context-dependent are [11,13]:

- the specific labor laws of each country,
- the different circumstances defining the resources and demands of each hospital,
- the scarcity/availability of nurses, and the nurse's credentials.

The performance and quality of any healthcare system depend, essentially, on the quality and motivation of hospital staff such as medical and paramedical personnel [14]. Nurse dissatisfaction that causes nurses frustration could lead to work under high stress and increase absenteeism and turnover [15]. So, a hospital needs to have an adequate planning tool for the assignment and scheduling of its human resources. In this paper, we solve a multi-objective nurse scheduling problem by using Lexicographic Goal Programming with dynamic Satisfaction Functions to assign nurses to different shifts taking into account the skills of nurses, their preferences, and absenteeism. A numerical example based on data from a real case study of the intensive care unit in Habib Bourguiba Hospital in Tunisia is used to illustrate the effectiveness of the incorporation of dynamic satisfaction function in the nurse-scheduling problem.

2. Literature review

The nurse scheduling problem is present in almost every healthcare institution. It is very important to nurses, patients, and head nurse managers. Several methods have been used to solve the nurse scheduling problem. Some researchers have relied on heuristics [10,16], others on meta-heuristics [7,17–21]. Some researchers have used hybrid methods such as [9,22–24]. Nowadays, thanks to computer and algorithmic improvement, exact methods have become promising [25]. For modeling the Nurse Scheduling Problem, we chose the technique of Goal Programming developed by Charnes and Cooper [26] to solve decision-making problems with multiple objectives. This method has the advantage of being very flexible, and it helps stimulate many variations in constraints and priority goals. It is an extension or a generalization of linear programming that addresses problems with conflicting goals. The Goal Programming model is based on the definition of aspiration levels or goals for each target while trying to find the closest solution to these goals [27]. This model aims to minimize the sum of positive and negative deviations from the objectives and goals to be met to obtain a satisfactory solution and a good compromise [27]. Goal Programming takes into account several objectives that may be in conflict, and the solution obtained by this procedure represents the best compromise that can be made by the decision-maker [28]. Among the variants of Goal Programming frequently used to solve the Nurse Scheduling Problem, we can cite standard Goal Programming, Weighted Goal Programming, and Lexicographic Goal Programming. Arthur and Ravindran [29] used the Goal Programming method in the first phase to assign the nurses to their day-on/day-off pattern for the two-week scheduling horizon. In the second phase, they used a heuristic procedure to assign working shifts to nurses. Nasir et al. [30] used the technique of Goal Programming to prepare a balanced schedule for 15 nurses in the coronary care unit of Shah Alam Hospital. They generated 15 schedules, which are used by each nurse for 15 days following a continuous rotation until all nurses used all schedules. Therefore, their method produced schedules covering 225 days. Azaiez and Al-Sharif [17] developed a binary linear Goal Programming model for nurse scheduling in Riyadh-Al-Kharj in Saudi Arabia hospital. They aimed to balance the workload based on the division of shifts per day and the satisfaction of the minimum number of hours worked per month, the maximum number of consecutive working days, and the minimum number of nurses. They also took into account nurses' preferences in terms of days off. Fitriani et al. [31] applied Fuzzy Goal Programming based on the MINMAX approach and they



$$F(\delta)\begin{cases} 1 & \text{if } \delta \le \alpha_d \\ \frac{\alpha_0-\delta}{\alpha_0-\alpha_d} & \text{if } \alpha_d < \delta \le \alpha_0 \\ 0 & \text{if } \alpha_0 \le \delta \le \alpha_v \end{cases}$$

Fig. 1. Continuous satisfaction functions.

set tolerance limits for each goal. Mohammadian et al. [32] identified nurses' preferences for days off and shifts through a questionnaire. They also ranked the goals according to their priority. Through the use of Goal programming, they generated a schedule for 27 nurses in the emergency department of a hospital in Tehran for 28 days. Jenal et al. [33] proposed a 0–1 Lexicographic Goal Programming to model a cyclical nurse schedule. Also Agyei et al. [34] applied a 0–1 Lexicographic Goal Programming to the outpatient department of Tafo Government in Kumasi, Ghana. Böóvarsdóttir et al. [35] solves the nurse scheduling problem using acceptance thresholds in the form of lexicographic goal programming. These thresholds are estimated automatically; they are specific to each instance and implicitly integrate compromises. Abdollahi and Ansari [36] used the Analytical Hierarchy Process and Weighted Goal Programming technique to model a schedule for nurses in an intensive care unit. The weights in the objective function of the Goal Programming model are obtained through the Analytical Hierarchy Process method. Wang et al. [37] generated a cyclical weekly schedule for ambulatory care nurses by using Weighted Goal Programming. Ariyani et al. [38] used Goal Programming to model a schedule for nurses who work in a special room for the treatment of COVID-19 inpatients at the Universitas Sebelas Maret hospital. They aimed to achieve the minimum number of male nurses and senior nurses in each shift and to ensure the same total number of working days for all nurses. Li et al. [39] proposed a hybrid approach of Goal Programming and metaheuristic search to find compromise solutions for the Nurse Scheduling Problem with several hard and soft constraints. They aimed to solve real-world Nurse Scheduling Problem instances in a Dutch hospital. Ang et al. [40] proposed a Lexicographic Goal Programming for an emergency department of a hospital in Singapore, which allowed the automatic generation of the nurses' schedule taking into account the managerial priorities and the nurses' preferences. Their model minimized the nurse–patient ratio deviation from 47% to 10%. Rerkjirattikal et al. [41] used a survey in an operating room in a hospital in Thailand to determine job satisfaction factors. They formulated a mathematical model using Goal Programming, to achieve equity and job satisfaction.

In all cited Goal Programming works, the decision-makers preferences (head nurse and nurses) were not taken into account in the same way as considered in this paper. We use the Goal Programming with dynamic Satisfaction Function more explicitly in solving the Nurse Scheduling Problem and meeting the head nurse requirements. The head nurse better manages the shortage and the surplus in the decision-making process and may revise the Satisfaction Functions that he has previously established [42]. Our model is based on discrete Satisfaction Functions as opposed to the work of Martel et al. [43] which considers only continuous deviation variables (Refer to Fig. 1). In their work, the type of continuous Satisfaction Functions is static for each objective. In our case, we use a nonstatic discrete Satisfaction Functions which depends on the nurse absenteeism.

Where $_d$ is the indifference threshold, $_0$ is the threshold of zero satisfaction and $_v$ is the veto threshold.

Nurse absenteeism has not been sufficiently taken into consideration by researchers. Recently Otero-Caicedo et al. [44] used a genetic algorithm to generate a monthly nursing schedule in a hospital in Colombia. To reduce the effect of unplanned nurse absenteeism, they

focus on two reactive policies, one to maintain the core schedule and another to privilege the exclusive use of substitute nurses. To compare these policies with a non-preventive policy, they used the Montecarlo simulation which showed the effectiveness of their preventive–reactive model.

In our work, we are interested in unplanned nurse absenteeism by integrating dynamic satisfaction functions to maintain the equitable distribution of nurses.

Our work includes the following three contributions:

- The integration of the dynamic aspect of the problem. For each situation, we define special satisfaction functions.
- The limitation of the deviations from the goals to evenly distribute the nurses' surplus or shortages.
- Among the limited discrete values of the deviations from the goals, we allow the decision maker to incorporate his preferences in the satisfaction functions.

When there are no absences in the Intensive Care Unit of Habib Bourguiba Hospital, the workload is fully covered. In this case, the head nurse prefers to distribute the surplus of the nurses equitably during the days of the week. Between 4 and 8 absences, it is no longer possible to have one or more surplus nurses during all the days of the week. In this case, the preferences of the head nurse change according to the new situation and the satisfaction functions change according to the new requirements of the head nurse. Not having a surplus has become tolerable by the head nurse. When the number of absent nurses reaches 9, the shortage is inevitable. In this case, new satisfaction functions appear to meet the new requirements of the head nurse. A shortage of one nurse has become acceptable. Lexicographic Goal Programming has the advantage of allowing ordering goals according to their importance. In our case, we consider the Lexicographic Goal Programming with the presence of Satisfaction Functions. All the different Satisfaction Functions are placed in different priority levels to maximize the decision maker's preferences. Our approach to solving the nurse scheduling problem has a managerial impact by incorporating the head nurse's preferences and assisting him in avoiding nurse shortages while taking into consideration the hospital policies.

3. Mathematical formulation of the problem

The Habib Bourguiba University Hospital in Tunisia has several problems that have been the focus of many researchers [45,46]. The mathematical model presented in this section is developed to solve the Nurse Scheduling Problem in the Intensive Care Unit of Habib Bourguiba University Hospital in Tunisia. The head nurse of this unit complained about several deficiencies:

- Organizational problems: nurses work according to rigid, unsatisfactory, and not rigorous schedules. This problem is mainly due to frequent and unplanned absenteeism.
- Information system problems: at the Intensive Care Unit there is a lack of communication between nurses.
  This problem is a major obstacle and a source of failure related to incomplete and imprecise instructions.
- Psychological problems: the rigidity of the schedule results in mental and physical fatigue for nurses.
- Planning problems: are due to the lack of computerized schedules.

The head nurse complained in particular about the unexpected and frequent absenteeism of nurses. This problem requires him to change the schedule already prepared which usually takes too much time. The use of Goal Programming will enable us to increase the level of flexibility, satisfy the desires of nurses in the case study of Habib Bourguiba Hospital Intensive Care Unit, and deal with frequent and unexpected absenteeism. The Nurse Scheduling Problem at the Habib Bourguiba Hospital specifically in the Intensive Care Unit is to assign a set of nurses to different shifts: morning, afternoon, night, and rest periods during a planning horizon in a full working week. The Intensive Care Unit has 42 nurses; they must be almost equally assigned to shifts. Table 1 provides the starting times, the finishing times, and the duration of the three shifts: morning, afternoon, and night. Table 2 presents the daily needs of a number of nurses for each shift.

Table 1
Types and duration of Intensive Care Unit at shifts.

	Time code	Start times	Finishing times	Net length
Morning	1	07H : 00	13H : 30	6h
Afternoon	2	13H : 00	19H : 30	6h
Night	3	19H : 00	07H : 00	12h

Table 2
Daily needs in nursing for each shift.

	From Monday to Saturday	Sunday
Morning	12	6
Afternoon	6	6
Night	6	6

Each type of shift has a daily requirement for nursing staff to cover the workload.

Table 2 shows that the workload is much higher in the morning compared to other shifts. This implies a higher need for 12 nurses, while on Sunday the Intensive Care Unit needs only 6 nurses for the whole day because it is a holiday. For the formulation of our problem, we need:

3.1. Input data sets

The following are the sets used in the model:

- N: all nurses in the Intensive Care Unit.
- D: all days covering the planning horizon from Monday to Sunday.
- D0: all days covering the planning horizon from Monday to Saturday.
- S: periods associated with each type of shift; S = {0, 1, 2, 3} which represents rest, morning, afternoon, and night shifts respectively.
- SZ: periods associated with shifts of the day and night; with SZ = {1, 2, 3} which represents the morning, afternoon, and night shifts respectively.
- SM: periods associated with shifts of the day; with SM = {1, 2} which represents the morning and afternoon shifts respectively.
- SAN: periods associated with shifts of the afternoon and night; with SAN = {2, 3} which represents the afternoon and night shifts respectively.
- T1: volunteer nurses who want to work only at night.
- T2: nurses who must work only during the day.
- T3: nurses who handle the dialysis machine.
- T4: the indifferent nurses who have no requirements and can work during any shift.

3.2. Decision variables

$$x_{nsd} = \begin{cases} 1 & \text{if the nurse n is assigned to the shift s on the day d} \\ 0 & \text{otherwise} \end{cases}$$

Where

- $n = 1; \ldots; 42$
- $s = 0; 1; 2; 3$
- $d = 1; \ldots; 7$

3.3. Deviation variables

The deviations variables δ^{+}_{ds} and δ^{-}_{ds} are integer:

- δ^{+}_{d1}; δ^{-}_{d1}: Positive and negative deviations from nurses' goal demand of the morning shift from Monday to Saturday.
- δ^{+}_{d2}; δ^{-}_{d2}: Positive and negative deviations from nurses' goal demand of the afternoon shift from Monday to Saturday.
- δ^{+}_{d3}; δ^{-}_{d3}: Positive and negative deviations from nurses' goal demand of the night shift from Monday to Saturday.
- δ^{+}_{7s}; δ^{-}_{7s}: Positive and negative deviations from nurses' goal demand throughout the Sunday shifts.

3.4. Defining the objective of the problem

The aim of solving the problem is to generate a schedule for the Intensive Care Unit nurses that respects the regulations of working time and the desires of the staff. This schedule should respect the equal distribution of workload throughout the shifts. To ensure the required number of nurses needed in the Intensive Care Unit, we have maximized the Satisfaction Functions which take into consideration the head nurse preferences.

We give priority to our goals according to their importance by adding p1 and p2 and our general objective function can be formulated as follows:

$$\text{Maximize } p1 * \sum_{d\in D}\sum_{s\in SZ} F2\ \delta^{-}_{ds} + p2 * \sum_{d\in D}\sum_{s\in SZ} F1(\delta^{-}_{ds}) + \sum_{d\in D}\sum_{s\in SZ} F1(\delta^{+}_{ds})$$

Where F1 and F2 are the Satisfaction Functions for regular nurses and competent nurses respectively. The satisfaction functions change dynamically according to the number of absent nurses. In our real case, there are three situations. The first one is when the number of absences is between 0 and 3 nurses, the second case is when the number of absences is between 4 and 8, and the last one is when the number of absences reaches 9 nurses. In each situation, the satisfaction functions change to meet the new requirements of the head nurse which explains the dynamic side of our work. In other hospitals, we can have different situations related to hospital regulations and local rules. In this case, it is possible to adapt the satisfaction functions to their regulations.

3.5. Identification of constraints

To develop a schedule certain rules must be respected. They come from regulations but also stems from agreements between manager and nurses. The constraints adopted in the case of the Intensive Care Unit of Habib Bourguiba Hospital are:

· Every nurse should be assigned to one and only one shift for each day of the planning horizon or is at rest.

$$\sum_{s\in S} x_{nsd} = 1 \qquad \forall\, n \in N \text{ and } d \in D \tag{1}$$

· Nurses must work 36 h per week whatever the type of shift.

$$\sum_{d\in D}\sum_{s\in SM} 6 * x_{nsd} + \sum_{d\in D} 12 * x_{n3d} = 36 \qquad \forall\, n \in N \tag{2}$$

· The nurse who works at night must necessarily have a day off tomorrow.

$$x_{n3d} + x_{nsd+1} \le 1 \qquad \forall\, n \in N,\ s \in SZ \text{ and } d \in D0 \tag{3}$$

· Nurses who work at night on Sunday must be at rest on Monday. In the case where the manager has no absenteeism and has the same volunteer nurses, he can use the same schedule.

$$x_{n37} + \sum_{s\in SZ} x_{ns1} \le 1 \qquad \forall\, n \in N \tag{4}$$

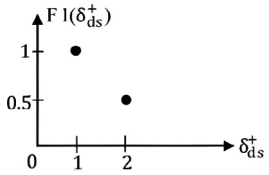


Fig. 2. Satisfaction Function of nurses surplus in the morning (case 0 to 3 absences).

· For Sunday and any type of shift, the Intensive Care Unit should have 6 nurses.

$$\sum_{n\in N} x_{ns7} + \delta^{-}_{7s} - \delta^{+}_{7s} = 6 \qquad \forall\, s \in SZ \tag{5}$$

· For each day from Monday to Saturday during the morning, the Intensive Care Unit should have 12 nurses.

$$\sum_{n\in N} x_{n1d} + \delta^{-}_{d1} - \delta^{+}_{d1} = 12 \qquad \forall\, d \in D0 \tag{6}$$

· For each day from Monday to Saturday during the afternoon, the Intensive Care Unit should have 6 nurses.

$$\sum_{n\in N} x_{n2d} + \delta^{-}_{d2} - \delta^{+}_{d2} = 6 \qquad \forall\, d \in D0 \tag{7}$$

· For each day from Monday to Saturday during the night shift, the Intensive Care Unit should have 6 nurses.

$$\sum_{n\in N} x_{n3d} + \delta^{-}_{d3} - \delta^{+}_{d3} = 6 \qquad \forall\, d \in D0 \tag{8}$$

· Each volunteer nurse should be assigned only to the night shift.

$$\sum_{d\in D0} x_{n3d} \ge 3 \qquad \forall\, n \in T1 \tag{9}$$

· Nurses having an age greater than 50 years or pregnant women must necessarily be assigned to the morning or afternoon shifts.

$$\sum_{s\in SM}\sum_{d\in D} x_{nsd} \ge 6 \qquad \forall\, n \in T2 \tag{10}$$

· For each day of the week, the Intensive Care Unit should have at least one competent nurse who can handle the dialysis machine.

$$\sum_{n\in T3}\sum_{s\in SZ} x_{nsd} \ge \text{limit} \qquad \forall\, d \in D \tag{11}$$

Where the limit takes the values 0, 1, 2, and 3 depending on the number of available competent nurses.

Our Nurse Scheduling Problem in the Intensive Care Unit is formulated as a Lexicographic Goal Programming with a dynamic Satisfaction Functions model. We treat 3 cases:

4 **Case 1 (from 0 to 3 absences)** the head nurse is well satisfied if there is a surplus of 1 nurse in the afternoon or the night. He is less satisfied if there is a surplus of 2 nurses. He prefers to have only a surplus of 1 nurse in the morning and avoid as much as possible a surplus of 2 nurses (0 satisfaction) because there are enough nurses to cover the workload in the morning (12 nurses as opposed to afternoon and night shifts which require only 6 nurses). Any solution given a shortage of competent nurses is rejected by the decision maker. For this case, we have chosen three Satisfaction Functions. A detailed schedule for this case is presented in Table 17 in Appendix B.

To satisfy the conditions imposed by each of the functions described in Figs. 2, 3, and 4, binary variables B_{ds} with the values 0 or 1 will be introduced. For the case of 0 to 3 absences, the Satisfaction Functions F1 and F2 are defined as follows:

$$F1(\delta^{+}_{ds}) = \begin{cases} 1 & \text{if } \delta^{+}_{ds} = 1 \quad \forall\, d \in D \text{ and } s \in SAN \\ 0{:}5 & \text{if } \delta^{+}_{ds} = 2 \quad \forall\, d \in D \text{ and } s \in SAN \end{cases}$$

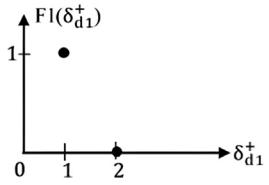


Fig. 3. Satisfaction Function of nurses' surplus in the afternoon and night shifts (case 0 to 3 absences).

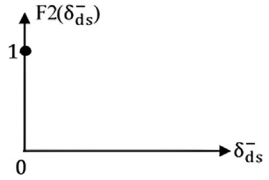


Fig. 4. Satisfaction Function of shortage of competent nurses (case 0 to 3 absences).

$$F1(\delta^+_{d1}) = \begin{cases} 1 & \text{if } \delta^+_{d1} = 1 \quad \forall\, d \in D \\ 0 & \text{if } \delta^+_{d1} = 2 \quad \forall\, d \in D \end{cases}$$

$$F2(\delta^-_{ds}) = 1 \quad \text{if } \delta^-_{ds} = 0 \quad \forall\, d \in D \text{ and } s \in SZ$$

We use the Lexicographic Goal Programming to give a high priority (p1) to the goal associated with the shortage of competent nurses. Wherein the model optimizes one goal at a time in each iteration based on priority. The optimal value of each goal becomes a bound for the next iteration. In our case, we resolve the goal associated to the shortage of competent nurses. The result generated is maintained and added to the constraints while optimizing the second goal related to the surplus of nurses.

Our objective function is as follows:

$$\text{Maximize} \sum_{d\in D}\sum_{s\in SZ} (B11_{ds} + 0{:}5 * B12_{ds} + B21_{d1} + p1 * B31_{ds})$$

We add the following constraints to the 11 constraints already mentioned.

· Constraints 12 and 13 ensure that we can only have a surplus of one or two nurses in the afternoon and night shifts.

$$B11_{ds} + 2 * B12_{ds} - \delta^+_{ds} = 0 \qquad \forall\, d \in D \text{ and } s \in SAN \qquad (12)$$

$$B11_{ds} + B12_{ds} = 1 \qquad \forall\, d \in D \text{ and } s \in SAN \qquad (13)$$

· Constraints 14 and 15 ensure that we can only have a surplus of one or two nurses in the morning shift.

$$B21_{d1} + 2 * B22_{d1} - \delta^+_{d1} = 0 \qquad \forall\, d \in D \qquad (14)$$

$$B21_{d1} + B22_{d1} = 1 \qquad \forall\, d \in D \qquad (15)$$

· Constraints 16 and 17 ensure that the shortage of competent nurses is equal to 0.

$$\delta^-_{ds} = 0 \qquad \forall\, d \in D \text{ and } s \in SZ \qquad (16)$$

$$B31_{ds} = 1 \qquad \forall\, d \in D \text{ and } s \in SZ \qquad (17)$$

$B11_{ds}$, $B12_{ds}$, $B21_{d1}$, $B22_{d1}$, and $B31_{ds}$ are binary variables.

4 **Case 2 (from 4 to 8 absences):** with at least 4 absences we cannot obtain a solution with a surplus of nurses in every shift. For this reason, the head nurse changes his Satisfaction Functions. He is more satisfied if there is a surplus of one nurse in the afternoon or night. In the morning shift the decision maker avoids the surplus of 1 nurse by assigning 0 satisfaction to this case (Fig. 5). Any solution giving a shortage of nurses or competent nurses is rejected. The Satisfaction Functions employed by the head nurse for this case change and are illustrated in Figs. 5, 6, 7, and 8. In this case, we give more importance to the Satisfaction Function related to the shortage of competent nurses followed by the shortage of regular nurses. This is obtained by assigning p1 and p2 to give priority to these Satisfaction Functions.

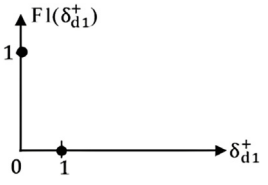


Fig. 5. Satisfaction Function of nurses' surplus in the morning shift (case 4 to 8 absences).

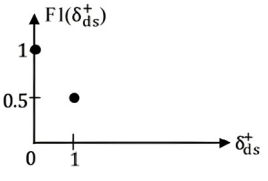


Fig. 6. Satisfaction Function of nurses surplus in the afternoon and night shifts (case 4 to 8 absences).

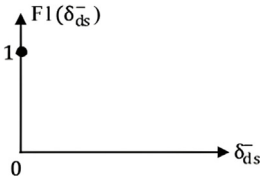


Fig. 7. Satisfaction Function of nurses shortages in all shifts (case 4 to 8 absences).

Fig. 8. Satisfaction Function of competent nurses shortages (case 4 to 8 absences).

$$F1(\delta^+_{d1}) = \begin{cases} 1 & \text{if } \delta^+_{d1} = 0\ \forall\, d \in D \\ 0 & \text{if } \delta^+_{d1} = 1\ \forall\, d \in D \end{cases}$$

$$F1(\delta^+_{ds}) = \begin{cases} 1 & \text{if } \delta^+_{ds} = 0\ \forall\, d \in D \text{ and } s \in SAN \\ 0{:}5 & \text{if } \delta^+_{ds} = 1\ \forall\, d \in D \text{ and } s \in SAN \end{cases}$$

$$F1(\delta^-_{ds}) = 1 \quad \text{if } \delta^-_{ds} = 0\ \forall\, d \in D \text{ and } s \in SZ$$

$$F2(\delta^-_{ds}) = 1 \quad \text{if } \delta^-_{ds} = 0\ \forall\, d \in D \text{ and } s \in SZ$$

Our objective function will be as follows:

$$\text{Maximize} \sum_{d\in D}\sum_{s\in SZ} (B11_{d1} + B21_{ds} + 0{:}5 * B22_{ds} + p2 * B31_{ds} + p1 * B41_{ds})$$

Subject to (all constraints except the 11 constraints already mentioned).

· Constraints 18 and 19 ensure that we can only have a surplus of zero or one nurse in the morning shift.

$$B12_{d1} - \delta^+_{d1} = 0 \qquad \forall\, d \in D \qquad (18)$$

$$B11_{d1} + B12_{d1} = 1 \qquad \forall\, d \in D \qquad (19)$$